

8 QUEENS ALGORITHM

AIM:

To place 8 queens on a standard 8×8 chessboard such that no two queens threaten each other (i.e., no two queens share the same row, column, or diagonal).

```
1  def print_board(board):
2      for row in board:
3          print(' '.join('Q' if x else '.' for x in row))
4      print()
5  def is_safe(board, row, col):
6      for i in range(row):
7          if board[i][col]:
8              return False
9      for i, j in zip(range(row-1, -1, -1), range(col-1, -1, -1)):
10         if board[i][j]:
11             return False
12     for i, j in zip(range(row-1, -1, -1), range(col+1, len(board))):
13         if board[i][j]:
14             return False
15     return True
16  def solve_queens(board, row):
17      if row == len(board):
18          print_board(board)
19          return True
20      for col in range(len(board)):
21          if is_safe(board, row, col):
22              board[row][col] = True
23              if solve_queens(board, row + 1):
24                  return True
25              board[row][col] = False
26      return False
27  def eight_queens():
28      board = [[False] * 8 for _ in range(8)]
29      if not solve_queens(board, 0):
30          print("No solution found")
31  eight_queens()
```

RESULT:

One or more valid arrangements of 8 queens on the board where none attack each other.